RESEARCH METHODOLOGY PRACTICAL

Experiment No. 1

Introduction to Research and Identification of Research Problem

Project Example: Skin Cancer Detection Using Deep Learning

Field	Details
Experiment No.	1
Title	Introduction to Research and Identification of Research Problem
Learning Outcome	Understand different types of research and formulate research problems.
Example Topic	Skin Cancer Detection Using Deep Learning

1. Experiment No.

Experiment No.: 1

2. Title of Experiment

Introduction to Research and Identification of Research Problem

3. Aim

To understand the concept, characteristics, and different types of research and to identify and formulate a suitable research problem in the field of computer applications.

4. Learning Outcome

After completing this experiment, the student will be able to:

- Understand the meaning and importance of research.
- Identify different types of research.
- Understand the concept of a research problem.
- Differentiate between a research topic and a research problem.
- Identify a suitable research problem from a broad research area.
- Formulate research objectives and research questions.

5. Introduction

Research is a systematic process of collecting, analyzing, and interpreting information to increase knowledge or solve a particular problem. It is an important part of computer science and information technology because new technologies, algorithms, and applications are continuously developed through research.

A research study normally begins with a broad area of interest. The researcher then identifies a specific problem or gap that requires investigation. A clearly defined research problem provides direction to the entire research process.

In this experiment, the research problem identification process is demonstrated using the project topic "Skin Cancer Detection Using Deep Learning."

6. Objectives

The main objectives of this experiment are:

1. To understand the basic concept of research.
2. To study the major types of research.
3. To understand the meaning and characteristics of a research problem.
4. To learn the steps involved in identifying a research problem.
5. To distinguish between a research topic and a research problem.
6. To formulate a research problem, objectives, and research questions using a practical example.

7. What is Research?

Research is a systematic and organized investigation conducted to discover new knowledge, verify existing knowledge, understand a problem, or develop a solution.

Research generally involves:

> *Problem Identification -> Literature Study -> Research Objectives -> Data Collection -> Analysis -> Interpretation -> Conclusion*

For example, in computer science, a researcher may investigate whether deep learning can improve the automatic classification of skin lesion images.

8. Characteristics of Research

The major characteristics of good research are:

1. Systematic

Research follows an organized sequence of steps rather than being based on random activities.

2. Objective

Research should be based on evidence and logical reasoning rather than personal opinions.

3. Empirical

Research often uses observations, experiments, measurements, datasets, or other forms of evidence.

4. Logical

The research process should follow proper reasoning and the conclusions should be supported by the findings.

5. Replicable

Where practical, the methodology should be described clearly enough that other researchers can repeat the study.

6. Problem-Oriented

Research generally focuses on answering a specific question or addressing a defined problem.

7. Analytical

Collected information is examined carefully to identify patterns, relationships, differences, or other meaningful findings.

8. Ethical

Research should be conducted responsibly, including appropriate treatment of data, participants, and research sources.

9. Types of Research

9.1 Basic Research

Basic research, also called fundamental research, is conducted mainly to increase knowledge and understanding of a subject without necessarily focusing on an immediate practical application.

Example: Studying how a particular deep learning architecture learns visual features from medical images.

9.2 Applied Research

Applied research focuses on solving a specific practical problem using existing knowledge or newly developed methods.

Example: Developing a deep learning system that assists in classifying skin lesion images.

9.3 Quantitative Research

Quantitative research uses numerical data and statistical methods to study a problem.

Example: Comparing the classification performance of different deep learning models using accuracy, precision, recall, and F1-score.

9.4 Qualitative Research

Qualitative research focuses on non-numerical information such as opinions, experiences, observations, or perceptions.

Example: Studying the experiences and opinions of healthcare professionals regarding the use of AI-based tools for assisting skin lesion assessment.

9.5 Exploratory Research

Exploratory research is conducted when a problem is not sufficiently understood. It helps the researcher explore possible causes, factors, approaches, or research directions.

Example: Exploring different deep learning techniques that could potentially be used for skin lesion classification.

9.6 Descriptive Research

Descriptive research describes the characteristics or existing conditions of a particular phenomenon.

Example: Analyzing the distribution of different categories of skin lesions in an available image dataset.

9.7 Experimental Research

Experimental research investigates the effect of one or more changes or conditions under controlled circumstances.

Example: Training two different deep learning models under comparable conditions and investigating whether a particular preprocessing or augmentation technique affects classification performance.

10. What is a Research Problem?

A research problem is a specific issue, difficulty, knowledge gap, or unanswered question that a researcher intends to investigate through systematic research.

A research problem should be sufficiently specific to guide the study.

Example

Research Topic: Skin Cancer Detection Using Deep Learning

Research Problem: Existing automated skin-lesion classification approaches may face challenges related to variations in lesion appearance, image quality, class imbalance, and generalization to images from different sources. Therefore, there is a need to investigate a suitable deep learning-based approach that can classify skin lesions reliably under realistic dataset conditions.

The second statement is a research problem because it identifies a specific issue that can be investigated.

11. Importance of Identifying a Research Problem

Identifying a clear research problem is important because:

1. It gives direction to the research.
2. It helps define the scope of the study.
3. It helps the researcher develop suitable objectives.
4. It determines what data needs to be collected.
5. It helps select appropriate research methods.
6. It prevents the study from becoming too broad.
7. It helps identify what contribution the research is expected to make.
8. It provides a basis for developing research questions.

A poorly defined research problem can result in unclear objectives, inappropriate methods, and weak conclusions.

12. Steps for Identification of a Research Problem

Step 1: Select a Broad Research Area

Choose a general area of interest. Example: Artificial Intelligence and Healthcare.

Step 2: Select a Specific Topic

Narrow the broad area to a particular subject. Example: Skin Cancer Detection Using Deep Learning.

Step 3: Conduct a Preliminary Literature Review

Study existing research papers, reports, datasets, and approaches to understand what has already been investigated.

Step 4: Identify Existing Problems or Gaps

Look for limitations, unanswered questions, inconsistent findings, or areas requiring further investigation.

Step 5: Check the Practical Importance

Determine why solving or investigating the problem matters.

Step 6: Narrow the Problem

Convert the broad issue into a specific and manageable research problem.

Step 7: Formulate the Problem Statement

Write the problem clearly and concisely.

Step 8: Develop Research Objectives

Define what the research intends to accomplish.

Step 9: Formulate Research Questions

Develop questions that the research will attempt to answer.

Step 10: Check Feasibility

Ensure that the required data, tools, time, computing resources, and skills are reasonably available.

13. Criteria for Selecting a Good Research Problem

1. Clear

The problem should be easy to understand and should avoid unnecessary ambiguity.

2. Specific

It should focus on a particular issue rather than a very broad subject.

3. Researchable

The problem should be capable of being investigated using appropriate research methods and available evidence.

4. Relevant

The problem should have academic, practical, or technological importance.

5. Feasible

The study should be achievable within the available time, resources, skills, and technology.

6. Original or Meaningfully Extending Existing Work

The research should provide a useful contribution, improvement, comparison, validation, or new perspective.

7. Ethical

The investigation should follow appropriate ethical principles.

14. Difference Between a Research Topic and Research Problem

Research Topic	Research Problem
A broad subject selected for investigation.	A specific issue or gap within the topic that requires investigation.
Usually describes the general area of study.	Explains what is wrong, unknown, difficult, or insufficiently studied.
May be only a few words.	Usually requires a clear explanatory statement.
Does not necessarily identify a gap.	Identifies a problem, limitation, or unanswered question.
Example: Skin Cancer Detection Using Deep Learning	Example: Investigating how a deep learning approach can improve reliable skin-lesion classification while addressing dataset and generalization challenges.

In simple words:

Topic = What you want to study.

Research Problem = What specific problem within that topic you want to investigate.

15. Example of Research Problem Identification

Project Topic: Skin Cancer Detection Using Deep Learning

A. Broad Research Area

Artificial Intelligence -> Machine Learning -> Deep Learning -> Computer Vision -> Medical Image Analysis

The broad research area is Artificial Intelligence in Healthcare, with a focus on computer vision and medical image analysis.

B. Specific Research Topic

Skin Cancer Detection Using Deep Learning

The topic focuses on applying deep learning and computer vision techniques to the analysis and classification of skin lesion images.

C. Existing Problem

Visual classification of skin lesions can be challenging because lesions may differ in:

- Shape
- Size
- Colour
- Texture
- Image quality
- Visual appearance

Automated image-based approaches can assist classification, but their performance may be affected by the quality and diversity of training data, class imbalance, preprocessing choices, model selection, and differences between training and unseen images.

D. Why the Problem is Important

Early and reliable identification of suspicious skin lesions can be valuable in medical decision-support systems. An automated model could potentially help organize or prioritize image-based assessments.

However, an AI model should not be treated as a replacement for professional medical diagnosis. The research is therefore focused on investigating the technical performance and limitations of an image-classification approach.

E. Limitations in Existing Approaches

Existing deep learning approaches may have limitations such as:

1. Dependence on the quality and diversity of datasets.
2. Imbalanced representation of different lesion categories.
3. Sensitivity to image preprocessing and augmentation choices.
4. Difficulty generalizing from one dataset or image source to another.
5. High performance on a particular dataset not necessarily guaranteeing equivalent performance in real-world clinical settings.

6. Limited interpretability of some deep learning predictions.

F. Formulated Research Problem

The research problem can therefore be formulated as:

> *To investigate a deep learning-based approach for skin lesion classification and evaluate how preprocessing, data augmentation, class distribution, model selection, and evaluation methodology affect its classification performance and generalization.*

16. Proposed Research Problem

Proposed Research Problem:

To develop and evaluate a deep learning-based image classification approach for skin lesion analysis, with particular attention to preprocessing, data augmentation, class imbalance, model selection, and the ability of the trained model to generalize to previously unseen images.

The study may use publicly available and appropriately licensed datasets suitable for research purposes.

17. Research Problem Statement

> *Skin lesion classification from dermoscopic or clinical images is a challenging computer vision problem because lesions can have considerable visual variation and image datasets may contain issues such as class imbalance and differences in image quality. Although deep learning models can learn visual patterns automatically, their performance can depend strongly on the dataset, preprocessing strategy, augmentation methods, model architecture, and evaluation procedure. Therefore, there is a need to systematically investigate and evaluate a suitable deep learning-based approach for skin lesion classification and examine its performance and limitations on unseen data.*

18. Research Objectives

The proposed research may have the following objectives:

1. To study existing deep learning and computer vision approaches used for skin lesion classification.
2. To prepare and preprocess a suitable skin lesion image dataset for deep learning-based classification.
3. To investigate the effect of preprocessing and data augmentation on model performance.
4. To train and compare suitable deep learning models for skin lesion classification using consistent evaluation procedures.
5. To evaluate the selected model using appropriate performance measures and analyze its limitations and generalization capability.

19. Research Questions

The proposed study may investigate the following questions:

1. How effectively can deep learning models classify different categories of skin lesions from images?
2. How does image preprocessing affect the classification performance of a deep learning model?
3. Does data augmentation improve classification performance, particularly when some classes have fewer training examples?

4. How do different suitable deep learning architectures compare under the same experimental conditions?
5. How well does the selected model perform on previously unseen images, and what limitations affect its generalization?

20. Expected Outcome

The expected outcome of the research is:

- A suitable dataset preparation and preprocessing pipeline for skin lesion images.
- A trained deep learning-based classification model.
- A comparison of selected models or approaches under consistent experimental conditions.
- An evaluation using appropriate classification metrics.
- An analysis of the effects of preprocessing, augmentation, and class distribution.
- Identification of the strengths and limitations of the proposed approach.
- A clearer understanding of whether the investigated approach can generalize beyond the training data.

The research should not assume beforehand that the proposed model will achieve a particular accuracy. The actual performance should be determined experimentally.

21. Conclusion

In this experiment, the basic concept of research and the importance of identifying a research problem were studied. Different types of research, including basic, applied, quantitative, qualitative, exploratory, descriptive, and experimental research, were discussed.

The difference between a research topic and a research problem was also identified. A topic represents the general subject of investigation, whereas a research problem describes a specific issue, limitation, knowledge gap, or unanswered question that can be investigated.

Using "Skin Cancer Detection Using Deep Learning" as an example, a broad research area was narrowed into a specific research problem. The existing challenges, importance, possible limitations, research objectives, research questions, and expected outcomes were identified.

Thus, a properly defined research problem provides a strong foundation for designing and conducting a systematic research study.

22. Viva Questions with Answers

Q1. What is research?

Answer: Research is a systematic investigation conducted to discover new knowledge, understand a problem, verify existing knowledge, or develop a solution.

Q2. What is a research problem?

Answer: A research problem is a specific issue, knowledge gap, difficulty, or unanswered question that a researcher intends to investigate.

Q3. What is the difference between a research topic and a research problem?

Answer: A research topic is the broad subject of study, while a research problem identifies a specific issue or gap within that subject that requires investigation.

Q4. Give an example of a research topic.

Answer: Skin Cancer Detection Using Deep Learning is an example of a research topic.

Q5. What is the research problem in the given example?

Answer: The research problem is to investigate how a deep learning-based approach can reliably classify skin lesions while considering challenges such as preprocessing, class imbalance, model selection, and generalization to unseen images.

Q6. What is basic research?

Answer: Basic research is conducted mainly to increase knowledge and understanding of a subject without necessarily focusing on an immediate practical application.

Q7. What is applied research?

Answer: Applied research focuses on solving a specific practical problem using knowledge and research methods.

Q8. What is quantitative research?

Answer: Quantitative research uses numerical data and statistical analysis to investigate a research problem.

Q9. What is qualitative research?

Answer: Qualitative research studies non-numerical information such as experiences, opinions, observations, and perceptions.

Q10. What is exploratory research?

Answer: Exploratory research is used to investigate a problem when there is limited understanding of the subject and to identify possible research directions.

Q11. What is descriptive research?

Answer: Descriptive research describes the characteristics or existing conditions of a particular phenomenon.

Q12. What is experimental research?

Answer: Experimental research investigates the effect of one or more variables or conditions under controlled circumstances.

Q13. Why is identifying a research problem important?

Answer: It gives direction to the research and helps determine the objectives, research questions, methodology, data requirements, and scope of the study.

Q14. What makes a research problem good?

Answer: A good research problem should be clear, specific, researchable, relevant, feasible, meaningful, and ethically appropriate.

Q15. What is a research objective?

Answer: A research objective is a clear statement of what the researcher intends to achieve through the study.

Q16. What is a research question?

Answer: A research question is a focused question that the study attempts to answer through systematic investigation.

Q17. Why is feasibility important while selecting a research problem?

Answer: Feasibility ensures that the study can be completed with available time, data, tools, skills, and resources.

Q18. What is meant by literature review?

Answer: A literature review is the study of existing books, papers, reports, and related work to understand what has already been done on a topic.

Q19. Why should AI-based medical tools be evaluated carefully?

Answer: They may be affected by data quality, bias, image differences, and generalization issues, so their limitations must be understood before practical use.

Q20. Can a deep learning model replace a doctor in skin cancer diagnosis?

Answer: No. In this practical example, the deep learning model is considered a decision-support or research tool, not a replacement for professional medical diagnosis.